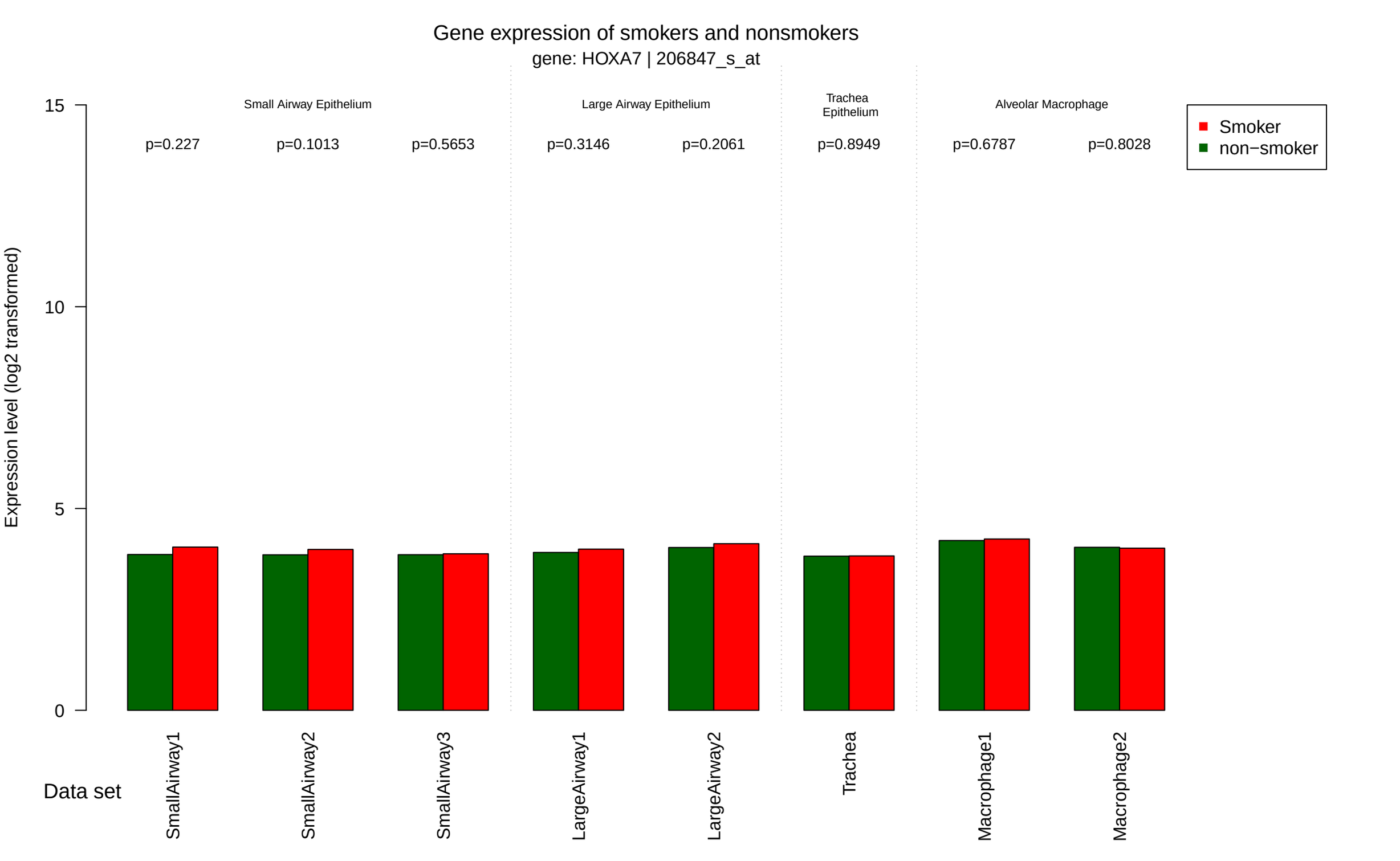
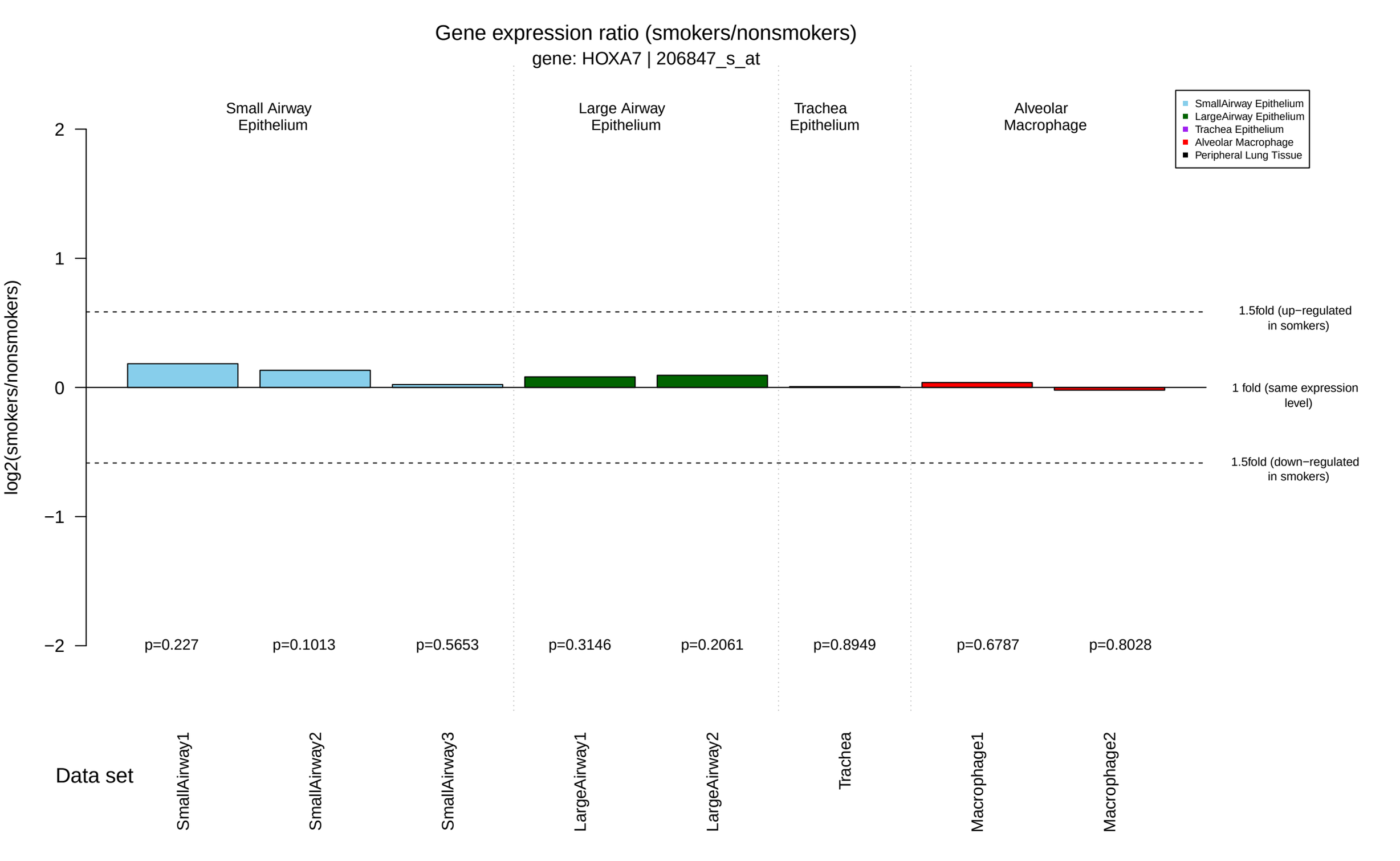
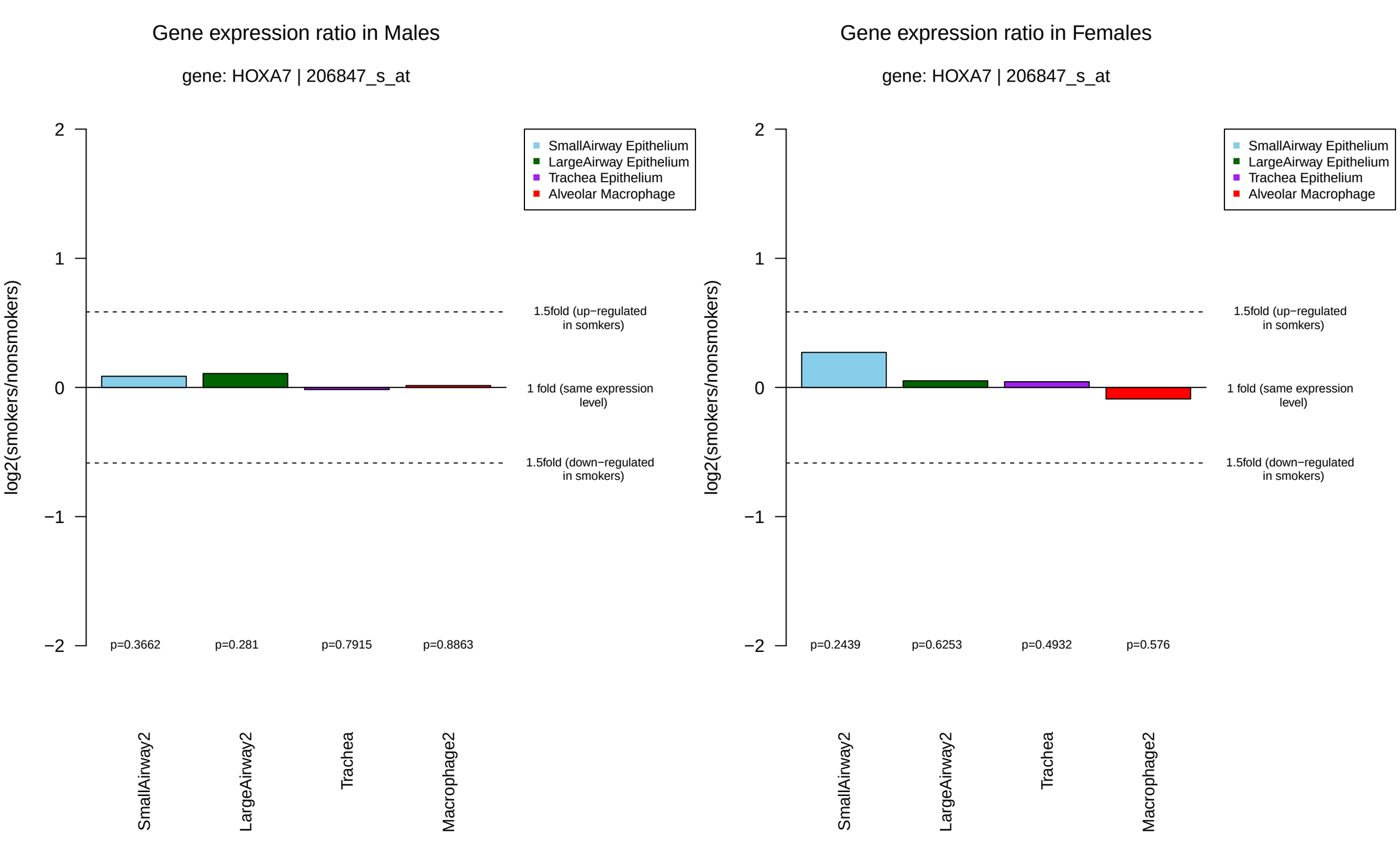
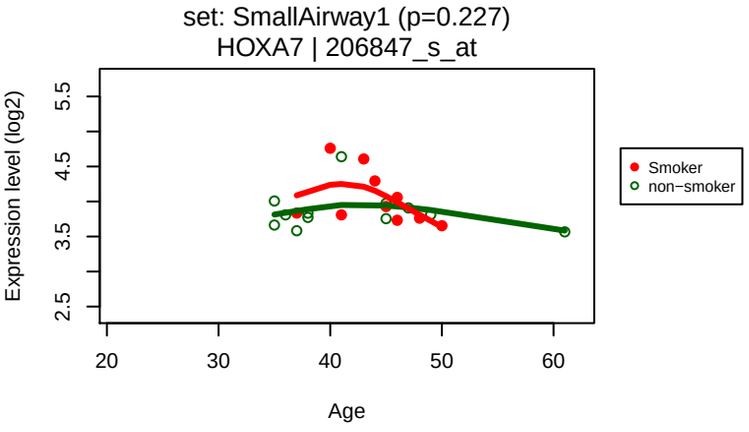
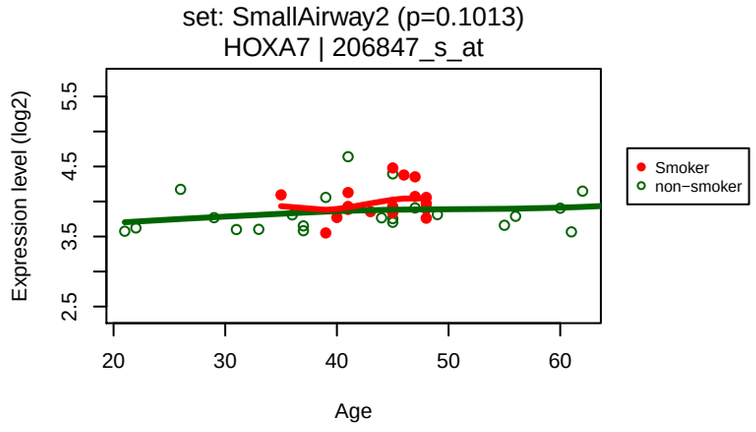
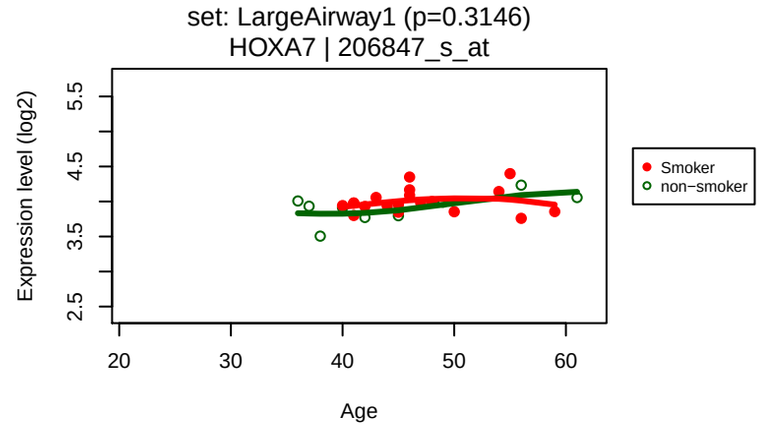

Gene expression of smokers and nonsmokers
gene: HOXA7 | 206847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.227
p=0.1013
p=0.5653
p=0.3146
p=0.2061
p=0.8949
p=0.6787
p=0.8028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

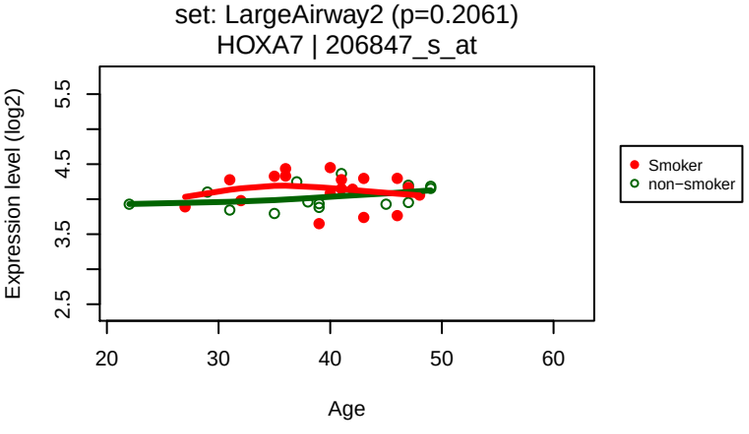

Gene expression ratio (smokers/nonsmokers)
gene: HOXA7 | 206847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.227
p=0.1013
p=0.5653
p=0.3146
p=0.2061
p=0.8949
p=0.6787
p=0.8028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

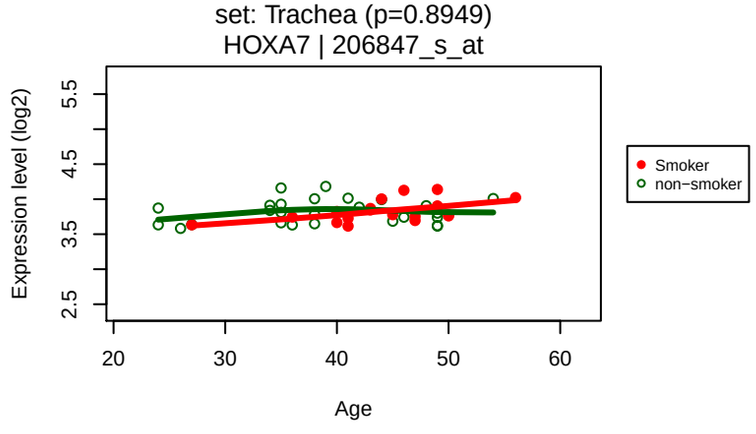

Gene expression ratio in Males
gene: HOXA7 | 206847_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3662
p=0.281
p=0.7915
p=0.8863
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA7 | 206847_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2439
p=0.6253
p=0.4932
p=0.576
SmallAirway2
LargeAirway2
Trachea
Macrophage2

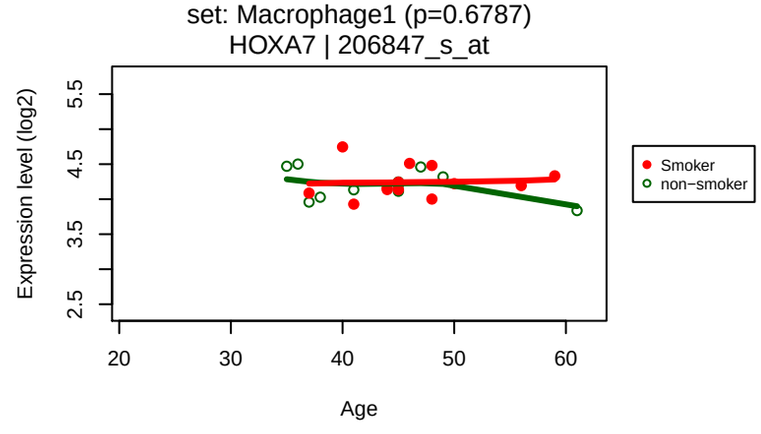

set: SmallAirway1 (p=0.227)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker

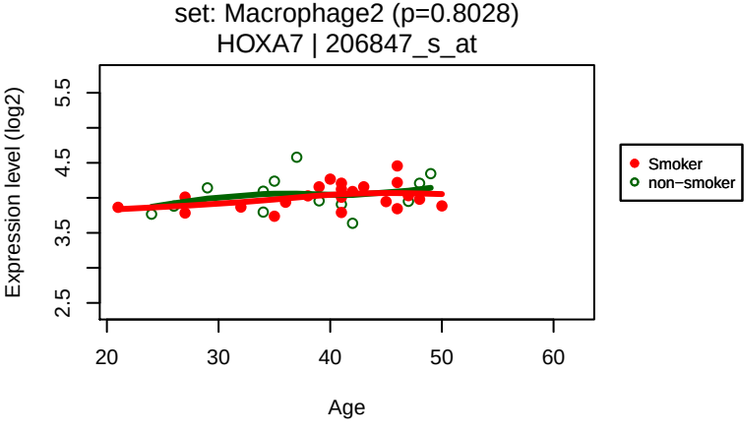

set: SmallAirway2 (p=0.1013)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker

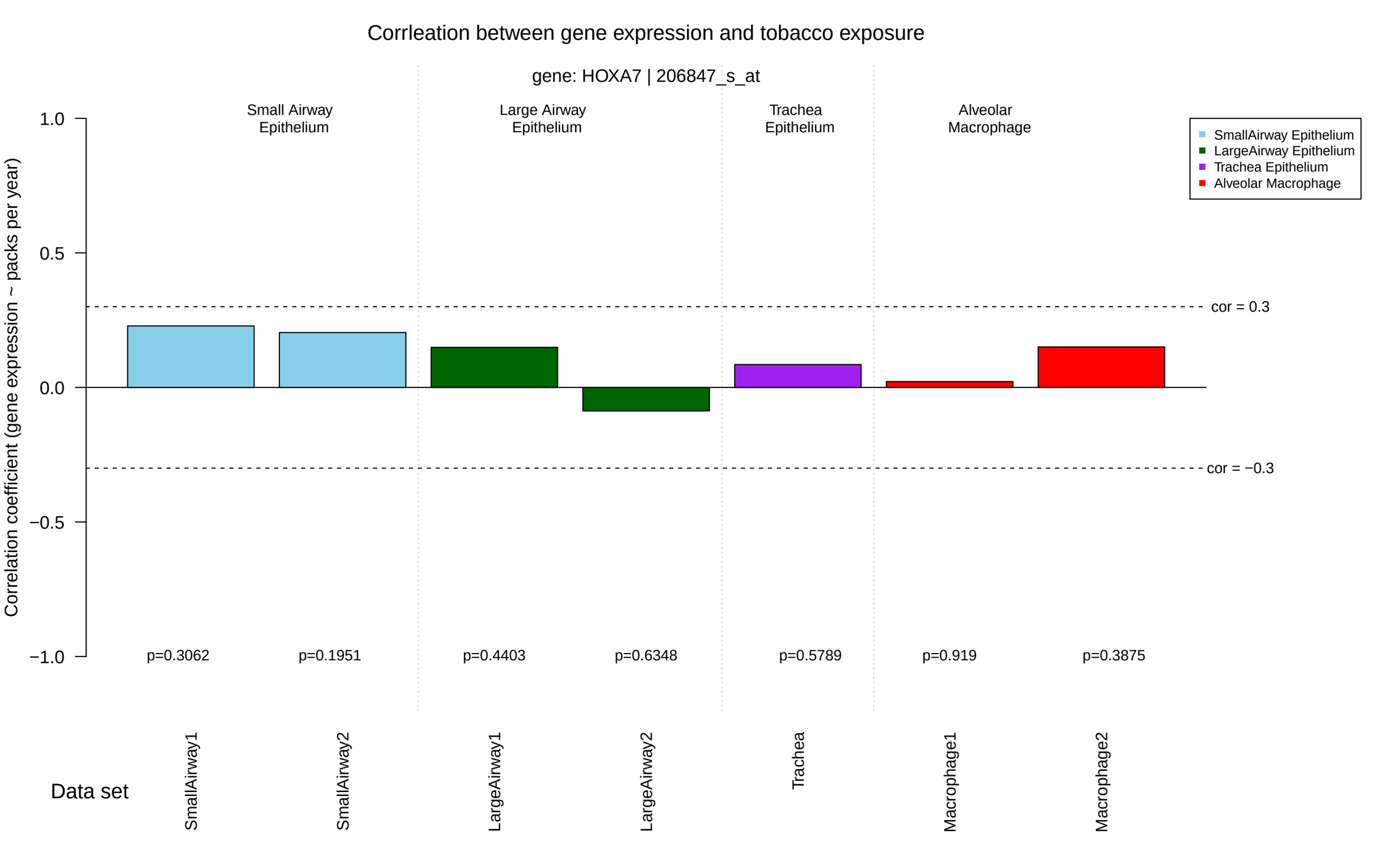

set: LargeAirway1 (p=0.3146)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2061)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8949)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6787)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8028)
HOXA7 | 206847_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA7 | 206847_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3062
p=0.1951
p=0.4403
p=0.6348
p=0.5789
p=0.919
p=0.3875
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2